University of Belgrade
Faculty of Organizational Sciences

BLOCKCHAIN LABORATORY

Prompt Engineering for Lovable

Second block · WHO BMI dataset workshop

From raw idea → polished prompt → working dashboard

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From idea to app

Use AI to clarify the request before Lovable builds it

What good prompting means in this workshop

- **Start from a health question,** not from a coding feature.
- **Use ChatGPT or Gemini as a** prompt architect before Lovable.
- **Ask for one visible change at a time,** then verify the result.

Think like a clinician. Prompt like a designer.

Always check if the chart, labels, and interpretation make sense. AI can still map data incorrectly.

Small, testable prompts beat one big rewrite.

Prompting workflow

1

Define the insight

What should the user learn from the WHO BMI data?

2

Polish the prompt

Ask ChatGPT or Gemini to expand, clarify, and structure it.

3

Build in Lovable

Paste the refined prompt and inspect the first result.

4

Refine with micro-prompts

Fix chart type, filters, labels, layout, or style one change at a time.

Example chain: "Help me rewrite this for Lovable" → "Create the dashboard" → "Change the chart type"

A reusable prompt structure for this dataset

Goal What change do you want?

Data Which dataset or columns matter?

View What should appear on screen?

Interaction What filters, clicks, or comparisons are needed?

Constraints What style or limits should Lovable follow?

Sample prompt

Using the WHO BMI dataset, create a clean and accessible dashboard for medical students. Show average BMI over time by sex, add a region filter and a year slider, and include a short insight box. Keep the design clinical, simple, and high-contrast.

Tip: mention real column names whenever possible.

Prompt makeover: vague → useful → iterative

Too vague

"Improve the BMI app."

Lovable-ready

"Using the WHO BMI dataset, add a line chart that compares average BMI over time by sex. Include a year range filter and keep the layout clean, clinical, and easy to read for medical students."

Then refine with micro-prompts

Replace the bar chart with a line chart.

Add a region dropdown at the top.

Shorten the interpretation text and improve contrast.

Best practice: use one prompt for structure, then smaller prompts for corrections.

Second block flow and team challenge

15 min
Prompting mini-lesson

20 min
Task briefing + Remix/download

30 min
Independent team work

20 min
Team presentations

Choose your challenge level

Basic

Change the visual style, improve the title, and add one useful chart.

Intermediate

Add a comparison, filter, or short interpretation for a public-health insight.

Advanced

Add a threshold, reference line, or guided feedback component.

In each presentation, show: your goal · your best prompt · the result · one lesson learned.

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